

W2 - OS Lab Work

1WA24CS241

1. Algorithm FCFS

Input: List of processes with (ProcessID, ArrivalTime, BurstTime)

Output: CompletionTime, TurnaroundTime, WaitingTime for each process

Code >>>

```
#include <stdio.h>
```

```
int main() {
```

```
    int n;
```

```
    int at[20];
```

```
    int bt[20];
```

```
    int wt[20];
```

```
    int tat[20];
```

```
    float twt = 0.0;
```

```
    float ttat = 0.0;
```

```
    float awt;
```

```
    float atat;
```

```
    printf("Stawan H. Ruiwale, 1WA24CS241\n\n");
```

```
    printf("Enter the num of process : ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter the arrival time for each process: ");
```

```
    for (int i = 0; i < n; i++) {
```

```
        printf("Process %d Arrival time : ", i + 1);
```

```
        scanf("%d", &at[i]);
```

```
    }
```

```
    printf("Enter the burst time for each process: ");
```

```
    for (int i = 0; i < n; i++) {
```

```
        printf("Process %d Burst time : ", i + 1);
```

```
        scanf("%d", &bt[i]);
```

```
    }
```

```
    // Calc wt and tat
```

```
    wt[0] = 0; //doesn't wait
```

```
    tat[0] = bt[0]; //tat for the 1st process is its bt
```

```
    for (int i = 1; i < n; i++) {
```

```
        //wt of current process = wt of previous process + bt of previous process
```

```
        wt[i] = wt[i - 1] + bt[i - 1];
```

```

        // tat = wt + bt
        tat[i] = wt[i] + bt[i];
    }

    // Calc total and avg times
    for (int i = 0; i < n; i++) {
        twt += wt[i];
        ttat += tat[i];
    }
    awt = twt / n;
    atat = ttat / n;

    // Display results
    printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");
    for (int i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n", i + 1, bt[i], wt[i], tat[i]);
    }

    printf("\nAverage Waiting Time: %.2f\n", awt);
    printf("Average Turnaround Time: %.2f\n", atat);

    return 0;
}

```

Output

>>>

```

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Enter the num of process : 4
Enter the arrival time for each process: Process 1 Arrival time : 0
Process 2 Arrival time : 0
Process 3 Arrival time : 0
Process 4 Arrival time : 0
Enter the burst time for each process: Process 1 Burst time : 7
Process 2 Burst time : 3
Process 3 Burst time : 4
Process 4 Burst time : 6

Process Burst Time      Waiting Time      Turnaround Time
P1       7              0              7
P2       3              7             10
P3       4             10             14
P4       6             14             20

Average Waiting Time: 7.75
Average Turnaround Time: 12.75

Process returned 0 (0x0)   execution time : 29.151 s
Press any key to continue.

```

2. Algorithm SJF_NonPreemptive :

Code

>>>

```
#include <stdio.h>

int main() {
    int n, i, j;
    int pid[20], at[20], bt[20], ct[20], tat[20], wt[20];
    int completed[20] = {0};
    int current_time = 0, completed_count = 0;
    int min_bt, index;

    printf("Stawan H. Ruiwale, 1WA24CS241\n\n");

    printf("Enter number of processes: ");
    scanf("%d", &n);

    for(i = 0; i < n; i++) {
        printf("Enter Arrival Time and Burst Time for P%d: ", i+1);
        scanf("%d %d", &at[i], &bt[i]);
        pid[i] = i + 1;
    }

    while(completed_count < n) {
        min_bt = 9999;
        index = -1;

        for(i = 0; i < n; i++) {
            if(at[i] <= current_time && completed[i] == 0) {
                if(bt[i] < min_bt) {
                    min_bt = bt[i];
                    index = i;
                }
            }
        }

        if(index == -1) {
            current_time++;
        } else {
            ct[index] = current_time + bt[index];
            tat[index] = ct[index] - at[index];
            wt[index] = tat[index] - bt[index];

            current_time = ct[index];
            completed[index] = 1;
            completed_count++;
        }
    }
}
```

```

    }
}

printf("\nPID\tAT\tBT\tCT\tTAT\tWT\n");
for(i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
        pid[i], at[i], bt[i], ct[i], tat[i], wt[i]);
}

return 0;
}

```

Output

>>>

```

C:\Users\Admin\Documents\ >
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Enter number of processes: 4
Enter Arrival Time and Burst Time for P1: 0 7
Enter Arrival Time and Burst Time for P2: 8 3
Enter Arrival Time and Burst Time for P3: 3 4
Enter Arrival Time and Burst Time for P4: 5 6

PID      AT      BT      CT      TAT      WT
P1        0        7        7        7        0
P2        8        3       14        6        3
P3        3        4       11        8        4
P4        5        6       20       15        9

Process returned 0 (0x0)   execution time : 30.220 s
Press any key to continue.
|

```

3. Algorithm SJF Preemptive :

Code

>>>

```
#include <stdio.h>

int main() {
    int n, i;
    int pid[20], at[20], bt[20], rt[20];
    int ct[20], tat[20], wt[20];
    int completed = 0, current_time = 0;
    int min_index, min_rt;

    printf("Stawan H. Ruiwale, 1WA24CS241\n\n");

    printf("Enter number of processes: ");
    scanf("%d", &n);

    for(i = 0; i < n; i++) {
        printf("Enter Arrival Time and Burst Time for P%d: ", i+1);
        scanf("%d %d", &at[i], &bt[i]);

        pid[i] = i + 1;
        rt[i] = bt[i];
    }

    while(completed < n) {
        min_rt = 9999;
        min_index = -1;

        for(i = 0; i < n; i++) {
            if(at[i] <= current_time && rt[i] > 0) {
                if(rt[i] < min_rt) {
                    min_rt = rt[i];
                    min_index = i;
                }
            }
        }

        if(min_index == -1) {
            current_time++;
        } else {
            rt[min_index]--;
            current_time++;

            if(rt[min_index] == 0) {
                completed++;
            }
        }
    }
}
```

```

        ct[min_index] = current_time;
        tat[min_index] = ct[min_index] - at[min_index];
        wt[min_index] = tat[min_index] - bt[min_index];
    }
}

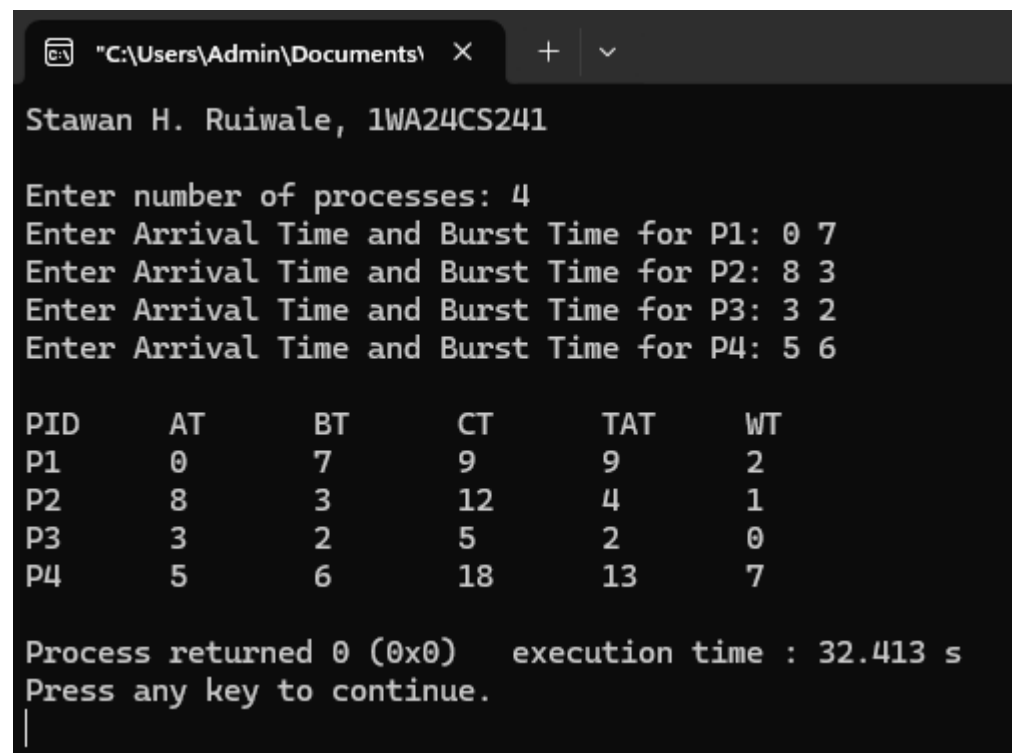
printf("\nPID\tAT\tBT\tCT\tTAT\tWT\n");
for(i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
        pid[i], at[i], bt[i], ct[i], tat[i], wt[i]);
}

return 0;
}

```

Output

>>>



```

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Enter number of processes: 4
Enter Arrival Time and Burst Time for P1: 0 7
Enter Arrival Time and Burst Time for P2: 8 3
Enter Arrival Time and Burst Time for P3: 3 2
Enter Arrival Time and Burst Time for P4: 5 6

PID      AT      BT      CT      TAT      WT
P1        0        7        9        9        2
P2        8        3       12        4        1
P3        3        2        5        2        0
P4        5        6       18       13        7

Process returned 0 (0x0)   execution time : 32.413 s
Press any key to continue.
|

```